3 Method

Interaction History → L0 Experiences → L1 Memory Cards → L2 Risk Rules

↓ Role-Conditioned Risk Retrieval and Evidence Budgets

Task Advocate || Risk Critic → Safety Decision Agent → Constrained Optimizer → Safety Guard → Execution

Fig. 1: Method overview. Hierarchical memory compresses long-term history while preserving provenance; each role receives distinct risk evidence; deterministic modules retain execution authority.

3.1 System Overview

We study high-level semantic action selection for an indoor mobile robot in a partially observable environment. At time t, the planning input is

$$\mathcal{X}_t = (g_t,\, r_t,\, e_t,\, \mathcal{A}_t), \tag{1}$$

where g_t denotes the task goal and progress, r_t the robot state, e_t the environment observation, and $\mathcal{A}_t$ the finite set of admissible actions. The system maintains history $\mathcal{H}_t$ and supplies task-relevant, traceable risk evidence within a bounded LLM context.

Figure 1 summarizes the pipeline. Executions form L0 risk experiences, which are compressed into L1 Risk Memory Cards; stable cross-experience patterns form L2 Risk Rules. The retriever builds separate evidence bundles for the Task Advocate, Risk Critic, and Safety Decision Agent. Each agent returns a structured recommendation and the experience and rule IDs it used. A deterministic constrained optimizer selects the action, and an independent Safety Guard verifies it.

Let $\mathcal{D}_t$ denote the three role reports and $\mathcal{Q}_t$ the matched L2 rules. The optimizer removes actions that violate hard constraints or L2 prohibition rules and solves

$$a_t^* = \arg\max_{a \in \mathcal{F}_t} U(a), \quad \text{s.t. } R(a) \le \rho,\ Z(a) \le \eta, \tag{2}$$

where $\mathcal{F}_t \subseteq \mathcal{A}_t$ is the feasible set, U is task utility, R is joint scene-and-memory risk, and Z combines observation confidence, memory coverage, and agent disagreement. Both are relative scores, not calibrated probabilities. Any conflicting L2 rules, invalid agent output, empty feasible set, or Safety Guard rejection triggers `safe_stop`, `observe_again`, or human review.

3.2 L0–L1–L2 Hierarchical Risk Memory

The hierarchy unifies risk experiences, bounds historical context, and preserves provenance from abstract rules to source cases. Its memory graph is

$$\mathcal{G}_t = (\mathcal{E}_t, \mathcal{M}_t, \mathcal{Q}_t, \mathcal{L}_t), \tag{3}$$

where $\mathcal{E}_t$, $\mathcal{M}_t$, and $\mathcal{Q}_t$ are the L0, L1, and L2 nodes, respectively, and $\mathcal{L}_t$ stores provenance links between stable IDs.

3.2.1 L0: Embodied Risk Experiences

L0 retains the interaction evidence required for auditing and re-abstraction. Experience i is

$$E_i = (G_i, S_i, A_i, O_i, F_i, K_i, D_i, P_i), \tag{4}$$

where G_i is the task goal, S_i the robot and environment state, A_i the executed action, O_i the outcome, F_i the failure cause, K_i the risk type and severity, D_i the decision record, and P_i the trajectory provenance and time. This structure links task, state, action, outcome, and risk cause for case reconstruction.

Each experience has an immutable `experience_id` and may link to sensor logs, control records, or human labels. Ingestion validates required fields, outcome values, severity ranges, and ID uniqueness. Agents receive compressed L1 representations, while L0 retains source evidence for compression and audit.

3.2.2 L1: Risk Memory Cards

L1 applies a deterministic compression map C:

$$M_i = C(E_i) = (T_i, H_i, A_i, O_i, F_i, B_i, Q_i, P_i), \tag{5}$$

where T_i is the trigger, H_i the hazard type and severity, B_i the mitigation and stopping conditions, Q_i the support count, reliability, and last validation time, and P_i the provenance. T_i retains only risk-relevant state, such as passage clearance, obstacle distance, observation confidence, and battery level.

Each card has a stable `memory_id` and stores its `experience_id` and source span. Agents reason over the card, while auditors follow $M_i \to E_i$ to the trajectory. The card also records its score decomposition and match rationale for reproducibility.

3.2.3 L2: Risk Rules and Cognitive Evolution

L2 represents stable patterns across related L1 cards as conditional rules:

$$Q_j = (C_j, K_j, A_j^+, A_j^-, B_j, N_j, c_j, z_j), \tag{6}$$

where C_j is the trigger, K_j the risk type, A_j^+ and A_j^- the recommended and prohibited actions, B_j the mitigation, N_j the source memory IDs, c_j the confidence, and $z_j \in \{\text{candidate}, \text{active}, \text{retired}\}$ the lifecycle state. A rule participates only when its trigger holds, and N_j provides the path $Q_j \to M_i \to E_i$.

A new trajectory creates an L0 experience; similar experiences update L1 support and reliability; repeated, sufficiently supported patterns create or strengthen candidate L2 rules. Contradictory evidence reduces confidence, revises the trigger range, or retires the rule. Minimum-support thresholds prevent a single failure from producing a strong rule.

At runtime, only *active* rules are matched. The system retains actions recommended by all matched rules and prohibited by none. If this intersection is empty, a deterministic conflict gate stops execution and requests human review.

For reproducible updates, each pattern maintains $n_j = (n_j^+, n_j^-, n_j^0)$, the counts of mitigation successes, risk failures, and human aborts. Without an explicit quality label, reliability is estimated as

$$c_j = \frac{n_j^+ + n_j^- + \alpha}{n_j^+ + n_j^- + n_j^0 + \alpha + \beta}, \tag{7}$$

where $\alpha, \beta > 0$ suppress extreme confidence from small samples. A candidate becomes *active* after meeting minimum provenance-count, cross-scene-consistency, and confidence thresholds. Persistent conflict, staleness, or counterexamples demote it to *candidate* or *retired*. Each transition creates a new version and retains the previous version and source IDs.

Failures identify prohibited actions and stopping conditions; successes validate mitigations. A rule extends to similar scenes only when both evidence types are represented, limiting conservatism caused by accumulated failures.

3.3 Role-Conditioned Risk-Aware Retrieval

Given $\mathcal{X}_t$ and role $r \in \{\text{adv}, \text{crit}, \text{dec}\}$, the retriever computes deterministic scene similarity from robot type, obstacle state, passage clearance, obstacle distance, observation confidence, and task goal. A positive-similarity candidate is scored by

$$s_i^{(r)} = \sum_{k \in \mathcal{K}} w_k^{(r)} \phi_{ik}, \quad \mathcal{K} = \{\text{sim}, \text{sev}, \text{rel}, \text{rec}, \text{fit}\}, \tag{8}$$

where $\phi_{ik} \in [0, 1]$ denotes scene similarity, risk severity, evidence reliability, recency, or role fit, and role weights $w_k^{(r)}$ sum to one. Reliability uses an explicit quality score when available and a smoothed posterior mean otherwise; recency decays with time since validation.

Role conditioning acts through $w_k^{(r)}$ and fit. The Advocate emphasizes successes and effective mitigations, the Critic failures, severe hazards, and stopping conditions, and the Decision role balances positive and negative cases. Thus, each role receives a different experience distribution.

Our auditable baseline uses weights ordered as $(\text{sim}, \text{sev}, \text{rel}, \text{rec}, \text{fit})$:

$$w^{(\text{adv})} = (.35, .05, .20, .05, .35), \tag{9}$$

$$w^{(\text{crit})} = (.30, .25, .15, .05, .25), \tag{10}$$

$$w^{(\text{dec})} = (.35, .20, .20, .10, .15). \tag{11}$$

These hand-set weights define a reproducible baseline. Ablation and sensitivity analyses isolate the effects of role fit, severity, and reliability.

To prevent near-duplicate cases from occupying Top-k, we use greedy diversity selection. Given selected set S, the next experience is

$$i^* = \arg\max_{i \notin S} \left[s_i^{(r)} - \lambda \max_{j \in S} d(i, j) \right], \tag{12}$$

where $d(i, j)$ measures redundancy in risk type, outcome, and mitigation, and λ controls the relevance-diversity trade-off. Selected experiences become L1 cards and are packed into $\mathcal{B}_r$ under budget B_r:

$$\sum_{M_i \in \mathcal{B}_r} \widehat{\tau}(M_i) \leq B_r, \tag{13}$$

where $\widehat{\tau}$ conservatively estimates tokens for bilingual JSON. Cards that exceed the budget are skipped, keeping context bounded as history grows.

The bundle contains the role, retrieval version, budget usage, L1 cards, score decomposition, match rationales, and source IDs, plus matched L2 rules. Agent outputs declare used evidence in `cited_experience_ids` and `cited_rule_ids`; IDs absent from the bundle are rejected. These records identify what was retrieved, why, and what the agent cited.

Algorithm 1 summarizes retrieval. For N experiences and at most K cards per role, feature computation costs $O(N)$ and greedy de-duplication $O(NK)$. Because the context budget fixes K, online cost is approximately linear in N. Roles share scene features but rank and pack separately.

Algorithm 1: Role-Conditioned Risk Retrieval
Input: state $\mathcal{X}_t$, role r, L0 store $\mathcal{E}$, budget B_r, limit K.
1) Validate the role and input schema;
2) Compute transparent scene similarity for each experience;
3) Compute severity, reliability, recency, and role fit;
4) Compute $s_i^{(r)}$ using $w^{(r)}$;
5) Greedily select candidates using the penalty in (12);
6) Deterministically compress candidates into L1 cards;
7) Pack cards under (13) and append matched L2 rules;
8) Return the bundle with score decomposition, match rationales, and source IDs.